

Quiz for Clean Kerala: A Statewide Water Quality & Sanitation Challenge

**For Higher Secondary Students
(+1, +2 of Kerala Syllabus , CBSE, ICSE)**

Preamble

Water quality and sanitation are among the most pressing concerns of our time. With the increasing incidence of water contamination and related issues, especially in regions like Kerala where traditional water sources such as wells and borewells are widely used, there is a growing need for awareness and education on these topics.

This programme is designed with the **objective of creating awareness among the student community about the importance of water quality and sanitation**. Recognizing the role of youth in driving sustainable change, the initiative seeks to equip students with the knowledge and tools necessary to understand and address water-related challenges in their surroundings. We are also prepared to provide study materials and facilitate interactive sessions for students who are interested in taking up assignments or projects related to water quality and sanitation. This awareness programme is a collaborative effort jointly organized by the **Centre for Water Research and Education (CWRE), Unnat Bharath Abhiyan (UBA), and the Indian Society for Technical Education (ISTE)**.

The CWRE, functioning under the **Department of Civil Engineering at Jyothi Engineering College**, is actively engaged in research, education, and outreach activities focused on water and sanitation. Our core mission includes **Information, Education, and Communication (IEC)** in the water sector, aimed at empowering communities to address water-related challenges. We operate a government-approved commercial water testing laboratory and offer water testing services at a nominal fee. In addition, CWRE conducts training programmes, publishes books and articles, and undertakes various community-oriented initiatives related to water conservation, quality monitoring, and sanitation awareness. We hope that this initiative will inspire students to become proactive contributors in promoting safe water practices and sustainable sanitation solutions in their communities.

Modality and Eligibility for the Water Quality & Sanitation Awareness Examination

- **Eligibility Criteria:**
 - Students currently studying in Plus One (+1) and Plus Two (+2) under the Kerala State Syllabus, CBSE, or ICSE are eligible to participate.
 - Only students from Kerala state are permitted to appear for the examination.
 - Participants must be currently enrolled in school and must produce a valid identity proof at the time of the examination.
- **Examination Details:**
 - The exam will be conducted in offline mode at Jyothi Engineering College. Students are required to make their own travel arrangements.
 - The question paper will consist of 100 multiple choice questions (MCQs).
 - Out of these, certain questions will be starred and used as tie-breakers if needed.
 - 60 sample questions are provided in advance, and 50 of these will appear in the final exam.

- The exam key will be published within 5 hours of completion of the examination.
- A formal challenge window will be provided for students to contest the answer key, within a specified time frame.
- **Registration and Fee:**
 - An **entrance fee of ₹100** will be charged per candidate.
- **Awards and Prizes:**
 - Attractive cash prizes will be awarded to the top performers:
 - **First Prize: ₹10,000**
 - **Second Prize: ₹7,000**
 - **Third Prize: ₹5,000**
 - All prize winners will also receive Certificates of Merit.
 - Those who score 50% or above will receive Consolation Certificate
- **Additional Information:**
 - Students may contact the Centre for Water Research and Education (CWRE) for any clarification or assistance.
 - In case of any disputes, the decision of the Administrative Committee will be final.
 - Further details will be made available through the official CWRE website
 - [cwre.jecc.ac.in](<http://cwre.jecc.ac.in>)
 - Tentative date for the written exam is 1st November .2025

Office bearers for the programme are as follows:

Sl.No	Name and Designation	Phone number
1	Fr. David Nettikkaden , Executive Manager	
2	Dr. P Sojanlal, Principal	
3	Dr. Alwyn Varghese, HOD CE	
4	Prof. Cyriac.M.G, General Convener	9447732619
5	Ms. Anna Joseph, Convener	9400947041
6	Mr. Suhas Nair S, Convener	6381153948
7	Ms. Jeffy Johny, Convener	7025168862
8	Mr. Alfred George, Convener	9995890881

(NB: The faculties mentioned in 4,5,6,7,8 may be contacted for any clarifications)

Syllabus

Topic: Water Quality and Sanitation Awareness

This syllabus is designed to provide students with a fundamental understanding of water quality, sanitation, and related public health and environmental concerns. The content is aimed at creating awareness and promoting informed action among young learners.

1. Water Quality and Sanitation: An Overview

- Global scenario of water quality and sanitation

- Relevance of water quality in daily life and health

2. Water Quality Parameters

- Physical Parameters:
 - pH
 - Turbidity
 - Conductivity
 - Total Dissolved Solids (TDS)
- Chemical Parameters
 - Chloride
 - Alkalinity
 - Hardness
 - Nitrate
 - Sulphate
 - Chemical Oxygen Demand (COD)
 - Biochemical Oxygen Demand (BOD)
- Biological Parameters:
 - Coliform bacteria
 - E. coli contamination

3. Contamination and Pollution

- Heavy metal contamination in water
- Pesticide pollution and its effects
- Sources and effects of contamination

4. Waterborne and Water-based Diseases

- Common diseases:
 - Cholera
 - Typhoid
 - Leptospirosis
 - Other bacterial and viral infections

5. Water Quality Issues in Kerala

- Local challenges in water quality
- Community-based and technological remedies

6. Water Treatment Methods

- Filtration
- Disinfection and Chlorination
- Use of water treatment kits and basic household practices

7. Sanitation and Waste Management

- Household Sanitation Systems:
 - Septic tank
 - Leach pit
 - Soak pit
- Solid Waste Management at Household Level

- Role of Local Self Government Departments (LSGD) in managing solid and liquid waste
- Personal and environmental hygiene
- Maintenance of home and surroundings
- Importance of regular hand washing

8. National and Global Initiatives

- Swachh Bharat Abhiyan (Clean India Mission)
- Sustainable Development Goals (SDGs) on water and sanitation

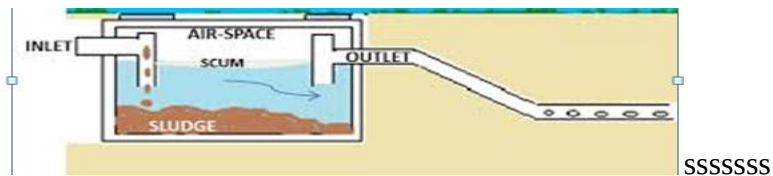
This syllabus form the basis for the multiple-choice questions (MCQs) in the examination. Students are encouraged to explore each topic for both theoretical understanding and practical relevance.

For any clarification or study support, students may contact the **Centre for Water Research and Education (CWRE)**.

Questions for the exam

Out of the 60 questions given below 50 will appear in the final exam , another 50 % will be added separately

1.0 What does the below picture indicate



a)leach pit b) septic tank c)Soak pit d)none

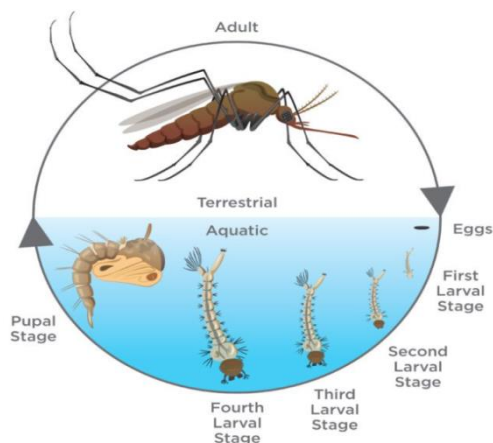
2.0 What does the below picture indicate



a)Hand washing is important for hygiene b) Hand washing prevents from diseases

c) children are more vulnerable for related diseases d) all of above

3.0 What does the below picture warn



a)Mosquito growth is dangerous for human life b) always cautious about mosquito growth c)stagnated water should be avoided around the premises d) all of the above

4.0 Which statement is appropriate for the given picture



- a. Blue baby syndrome b)keratosis c) fluorosis d) none

5) What does the picture below indicate



- a. Fluorosis b)keratosis c) Blue baby syndrome d) none

6) What does the below picture indicate and used for what



- a. Leach pit for excrete disposal b) kitchen pit kitchen waste c) soak pit for disposal of waste water d) none

7)Picture below is Dimitri Ivanosky, a Russian scientist , He is famous for what



- a. Discovery of cholera b) discovery of virus c) invention of antibiotic d) none

8. Hardness is caused by

a) Calcium b) Magnesium c) both Calcium and Magnesium d) Multivalent metallic cations

9.0 SDG refers to the step by which organization

a) World bank b) UN c) USA d) WHO

10.0 Which of the following is a waterborne disease?

(a) Measles (b) Typhoid (c) Chickenpox (d) Flu

11) Water Chlorination is essential as it removes ----- from water

a) Pathogenic organisms b) turbidity c) hardness d) none

12.0 What does "WASH" stand for in the context of water, sanitation, and hygiene?

(a) Water and Sanitation for Health (b) Water, Air, and Sanitation for Humans (c) Water, Sanitation, and Hygiene (d) Water and Sanitation for Healthcare

13.0 Which of the following is a consequence of poor sanitation?

(a) Reduced disease transmission (b) Improved public health (c) Increased spread of infectious diseases (d) Cleaner environments

Answer: (c) Increased spread of infectious diseases

14.0 Which of the following is a benefit of hand washing with soap?

(a) Reduced risk of respiratory infections (b) Increased risk of foodborne illnesses (c) Improved oral hygiene (d) No impact on health

15.0 Black colour of water indicate

a) good quality b) organic pollution c) bacterial presence d) Fluoride pollution

16.0 Name the scientist whose belief in god influenced scientific perspectives of his research

a) Henry cavendish b) ioseph aspidin c) Louis paster d) Ronald ross

17.0 Activated carbon which is highly useful in water treatment is obtained from

a) coconut shells b) waste wood c) red earth d) none

18) What is the purpose of the clarification or separation step at a wastewater treatment plant?

(a) To remove large solid objects (b) To separate scum and sludge from the liquid wastewater

(c) To treat wastewater with chlorine or ultraviolet light (d) To mix liquid wastewater with air to help bacteria break down solids

19. Which of the following is a consequence of poor sanitation?

a. Reduced disease transmission (b) Improved public health (c) Increased spread of infectious diseases (d) Cleaner environments

20. What is the primary purpose of a wastewater treatment plant?

(a) To make water potable (b) To remove pollutants from wastewater (c) To increase water levels

(d) To generate electricity

21) Which of the following is a waterborne disease?

- (a) Malaria (b) Tuberculosis (c) Cholera (d) Dengue

22) What does BOD in water quality testing stand for?

- (a) Basic Oxygen Demand (b) Biological Oxygen Demand (c) Bacterial Oxygen Demand (d) Biochemical . 23. Acceptable standard as per IS 10500 2012 for sulphate in mg/litre

- A) 400 b) 200 c) 250 d) 450

24 . Permissible limit for hardness IS 10500 2012 in mg/litre

- A) 500 b) 600 c) 700 d) 650

25. Drinking water quality standard in India is proclaimed by

- A) Central Pollution control board b) BIS c) NEERI Nagpur d) CWPR Oxygen Demand

26) White deposit on boiling in water indicates presence of

- A) Iron b) hardness c) chloride d) nitrate

27). What is the important health significance of hardness?

- A) on boiling white deposit is deposited b) hard water is good for heart diseases c) lather is formed with soap d) scaling in boilers

28. When chlorine gas is added to water pH will

- a) increase b) decrease c) remains same d) none

29. Which of the following is a method for conserving water?

- (a) maintain water sources effectively (b) Rainwater harvesting (c) Using water-intensive appliances
(d) all of the above

30) What is the main goal of the Swachh Bharat Mission?

- (a) Promote tourism (b) Clean India (c) Improve infrastructure (d) Increase literacy

31) Cholera is a dreadful disease. Which bacteria are responsible?

- a) bacillus b) staphylococcus c) Vibrio d) none

32) When should children wash their hands?

- a) Before eating or preparing food b) After using the toilet or hands are visibly dirty
c) After coughing, sneezing, or blowing their nose d) All of the above

33) What is the role of a septic tank in sanitation?

- (a) Purifies water for drinking (b) Treats wastewater from households (c) Stores rainwater
(d) Filters air

34. What is the significance of chlorination in water treatment?

(a) Removes turbidity (b) Kills harmful bacteria(c) Adds minerals(d) Improves taste

35.What is the effective method for disposal of plastic at domestic level

a) shredding b) Reuse c)landfill d) none

36)Which of the following is true about bacteria?

a) Bacteria multiplies and grows faster in warm environments') Bacteria needs air to survive c). Every type of bacteria can give people food poisoning.

d) By freezing food you can kill bacteria.

37) Deep well disposal of wastes not preferred due to

a)It reaches ground water easily b) It reaches surface water easily c) both d) none

39)What element is added to water to prevent tooth decay?

a). Chlorine b). Fluoride). Sugar). None of these

40)Which agency handles global health issues?

a) The World Health Organization, WHO b) the Food and Agriculture Organization, FAO

c) the Centres for Disease Control d). the Food and Drug Administration, FDA

40) Eutrophication in rivers is caused by

a) discharge of sediment in to the river b) excessive discharge of nutrients c) discharge of chlorides e) discharge of toxic substances

41)World hand wash day is celebrated on

a) 15th october b) 15th September c) 15th November d) none

42) Say true or false

Water logged areas are more prone for contamination due to ingress of surface contaminants

a. True b) False

43.Which process is used to remove dirt and sand from wastewater?

a). Aeration b) Chlorination c) Sedimentation d) Flocculation

44. When acid is added to water pH

a) Increases b) decreases c) no change d) none

45.Which of the following is not a waterborne disease?

(a)Measles (b) Typhoid(c) Cholera(d) Hepatitis

46.Jala jeevan mission is a programme of Government of India to promote

a) health and sanitation b)water quality and sanitation c) Provide safe water to the public by improving water supply systems in the country d) None

47. What is the major advantage of water cycle

a) Natural treatment and distribution of water in the earth b) Conversion of water into clean water c) Water conveyance through the earth d) None

48. Which of this major illness is caused due to non-availability of sanitation facilities?

a. Fever b) Cold c) Diarrhoea d) none

49. Which of the following is a benefit of hand washing with soap?

(a) Reduced risk of respiratory infections (b) Increased risk of foodborne illnesses (c) Improved oral hygiene (d) No impact on health

50. What is meant by health sanitation linkage

a) Poor sanitation practices being a major contributor to disease and illness b) Health is directly linked to sanitation c) Sanitation measures are essential for all living beings

d) All of the above

51. Which ministry initiated the Swachh Bharat Mission?

A. Ministry of Environment and forest B. Ministry of Urban Development Housing
C. Ministry of Urban development D. Ministry of Drinking water and Sanitation

52. Who is the inspiration behind the concept of SBM?

A. Mahatma Gandhi B. Jawaharlal Nehru C. Sardar Patel D. Morarji Desai

53. Which award is given by the Indian Government to cities and villages for cleanliness?

A. Swachh Puraskar B. Swachhta Puraskar C. Clean Region Award D. Nirmal Puraskar

54. Swachh Bharat Abhiyan was launched on the occasion the birth anniversary of a famous Indian leader. Who was it?

A. Mahatma Gandhi B. Netaji Subhash Chandra Bose C. Sardar Patel

55. Who are responsible to keep neighbourhoods clean?

A. Municipal corporation B. Citizens and NGOs C. Municipal corporation, Citizens and NGOs

56. Inspired by the Prime Minister of India, which country launched a similar event as the SBM on 3rd January 2015 in his own country?

A. Nepal B. Bhutan C. Sri Lanka D. China

57. Why is hand washing necessary after defecation

A. For self-cleanliness & hygiene B. For home and family member's cleanliness C. Both a & b

58. Say true or false

Four SB Mission objectives are Elimination of open defecation, Eradication of manual scavenging and Capacity Building of ULB

59. pH value can be increased by adding

a. Carbonic acid b) sodium chloride c) lime d) none

60. Organic pollution in water is assessed by the presence of

a) Biochemical oxygen demand b) chemical oxygen demand C) both D)None